

A Mathematical Consulting Report on Chiral Line-Field Experiments

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Abstract

This note explains the chiral line-field experiment in the same order in which the code runs it. The data are short line segments extracted from images. Each segment gives a location and an orientation, but not an arrow direction, so angles separated by π are treated as the same observation. We first study the uncleaned and cleaned files, then compare two kinds of models: low-dimensional spiral formulae indexed by an exponent p , and non-parametric Gaussian-process/RKHS kernel regressions. The comparison uses two mathematically different scores: held-out axial RSS, which measures predictive error on unseen folds, and evidence-style scores, which combine fit with a complexity penalty through Gaussian-process marginal likelihood or BIC.

1 What the Data Are

The experiment reads the files `data/**/arrow_segments.mat`. Each row is a segment with endpoints (x_{1i}, y_{1i}) and (x_{2i}, y_{2i}) . The location used for modelling is the midpoint

$$z_i = (x_i, y_i) = \left(\frac{x_{1i} + x_{2i}}{2}, \frac{y_{1i} + y_{2i}}{2} \right). \quad (1)$$

The image y -axis points downward, so the code flips the vertical coordinate before forming the mathematical angle

$$\hat{\alpha}_i = \text{atan2}\{-(y_{2i} - y_{1i}), x_{2i} - x_{1i}\}. \quad (2)$$

This is an *axial* observation. A line segment has no head and no tail, so $\hat{\alpha}_i$ and $\hat{\alpha}_i + \pi$ describe the same observed line. Directional statistics handles this by doubling angles, because α and $\alpha + \pi$ become 2α and $2\alpha + 2\pi$, which are identical on the circle. This is the standard axial-data construction described in Mardia and Jupp, *Directional Statistics*, Chapter 1, Section 1.1 [3].

The cleaned experiment uses a central circular region of interest. If `arrow_maps.mat` has shape 150×150 , the centre is $(75, 75)$ and the radius is $0.39 \cdot 150 = 58.5$. A segment is retained exactly when

$$\|z_i - (75, 75)\|_2 \leq 58.5. \quad (3)$$

This rule is not a theorem; it is an experimental preprocessing rule chosen to remove the peripheral line fragments and keep the central circular line field. Its effect is measured directly: across the 39 datasets, the median retained fraction was 0.992, the retained fraction ranged from 0.414 to 1.000, and the median cleaned sample size was 123 segments.

2 Data Study: Uncleaned and Cleaned Files

Let the raw dataset for file j be

$$\mathcal{D}_j^{\text{raw}} = \{(z_{ij}, \hat{\alpha}_{ij}) : i = 1, \dots, n_j^{\text{raw}}\}.$$

The cleaned dataset is the deterministic subset

$$\mathcal{D}_j^{\text{clean}} = \{(z_{ij}, \hat{\alpha}_{ij}) \in \mathcal{D}_j^{\text{raw}} : \|z_{ij} - c_j\|_2 \leq R_j\}, \quad n_j^{\text{clean}} \leq n_j^{\text{raw}}.$$

For the available arrow maps, $c_j = (75, 75)$ and $R_j = 58.5$. This means the cleaning rule changes only the index set of retained observations; it does not rotate, smooth, or re-estimate any angle. The raw files contain 5,092 total segments across 39 datasets. The cleaned files contain 4,721 retained segments, so 371 segments are removed. On a per-dataset basis, n_j^{raw} has mean 130.56 and median 128, while n_j^{clean} has mean 121.05 and median 123. The median retention ratio $n_j^{\text{clean}}/n_j^{\text{raw}}$ is 0.992, which shows that most datasets are changed lightly, but a few peripheral-heavy files are changed strongly.

Uncleaned vs cleaned data study across 39 line-field datasets

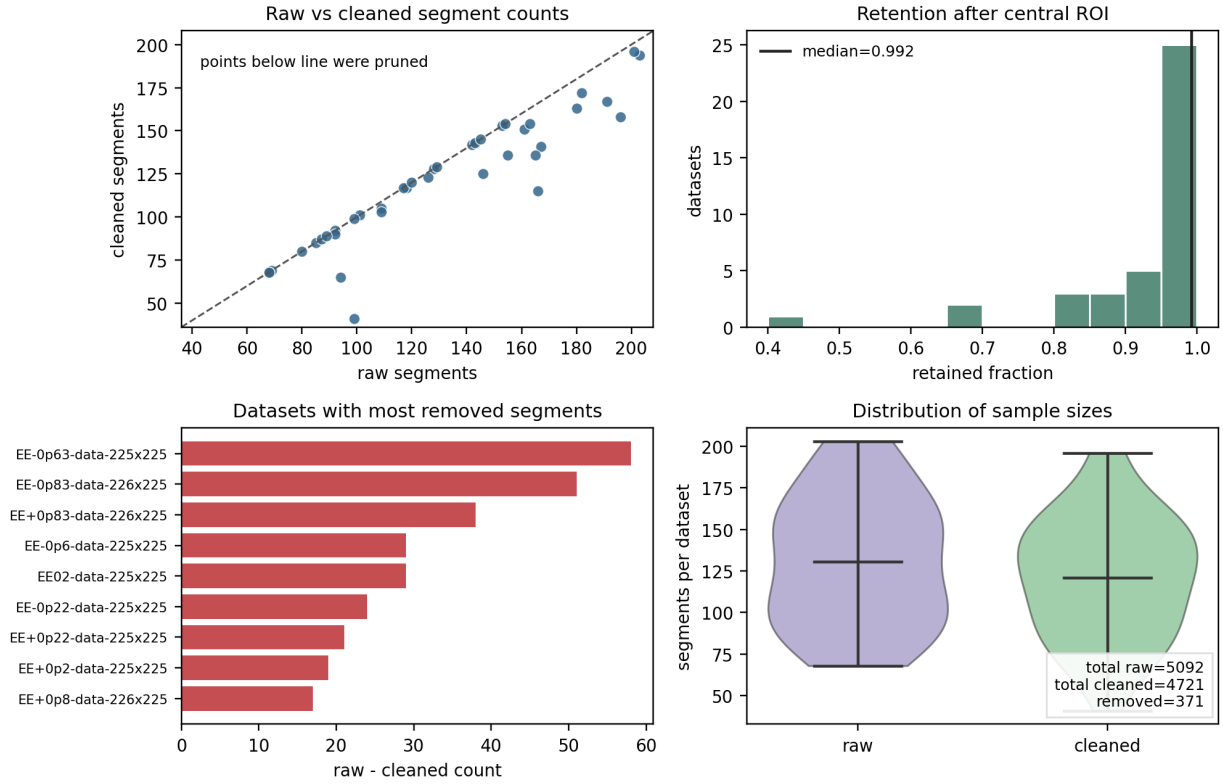


Figure 1: Quantitative data study comparing the uncleaned and cleaned files. The top-left panel plots n_j^{raw} against n_j^{clean} ; points on the dashed line would mean no pruning. The top-right panel shows the retention ratio. The bottom-left panel identifies the datasets for which the central ROI removes the most segments. The bottom-right panel compares the raw and cleaned sample-size distributions.

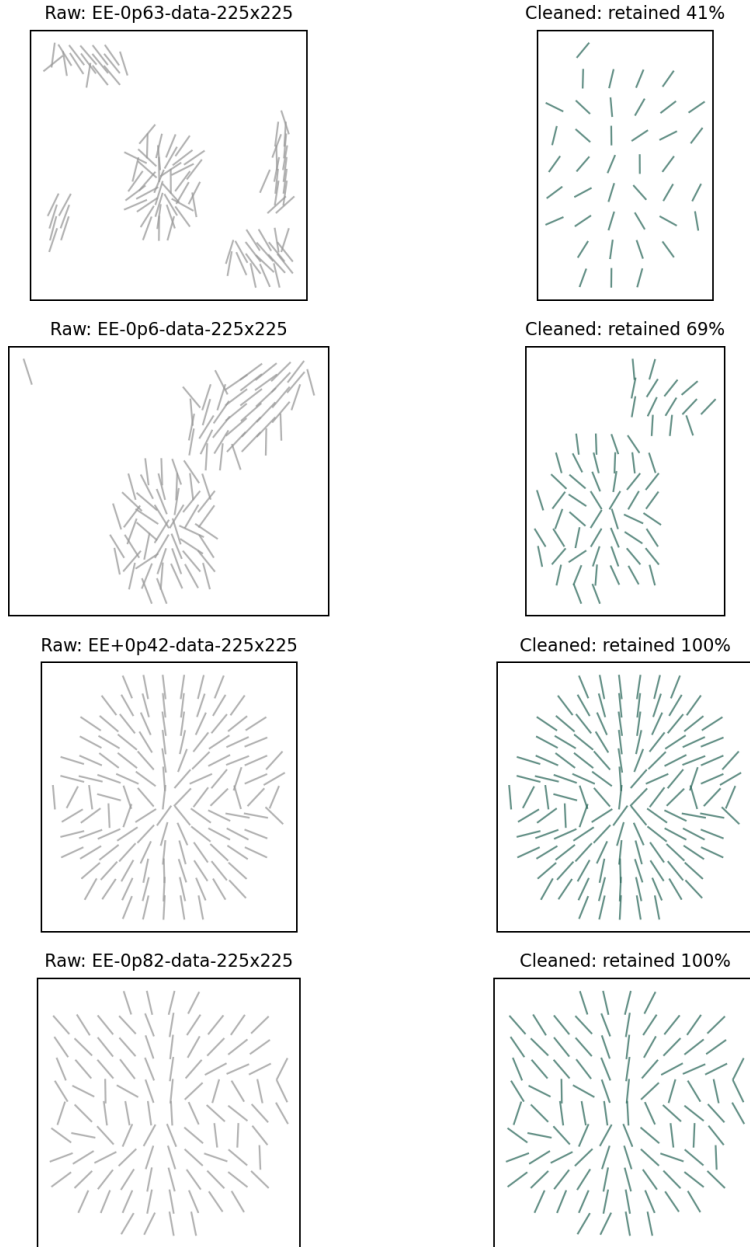


Figure 2: Representative raw and cleaned line fields. The left column shows the uncleaned segment set, and the right column shows the central-ROI subset used in the model comparison. These plots visually justify the data-design statement that cleaning removes peripheral fragments while retaining the central circular line field.

3 The Error for a Line Field

A signed angular error $\hat{\alpha}_i - \alpha_i$ would be wrong for these data because it would treat two equivalent line directions, α and $\alpha + \pi$, as different. The experiment therefore uses the axial residual

$$\Delta_i^{\text{ax}}(\mathcal{M}) = \frac{1}{2} \text{atan2}(\sin\{2[\hat{\alpha}_i - \hat{\alpha}_{\mathcal{M}}(z_i)]\}, \cos\{2[\hat{\alpha}_i - \hat{\alpha}_{\mathcal{M}}(z_i)]\}). \quad (4)$$

The factor of two removes the axial ambiguity, and the final factor of $1/2$ maps the residual back to an angle in $[-\pi/2, \pi/2]$. A perfect fit has $\Delta_i^{\text{ax}} = 0$. A model that is perpendicular to the observed line has $|\Delta_i^{\text{ax}}| = \pi/2$.

For prediction, the code reports held-out axial RSS,

$$\text{RSS}_{\text{test}}(\mathcal{M}) = \frac{1}{K} \sum_{k=1}^K \frac{1}{|I_k|} \sum_{i \in I_k} \{\Delta_i^{\text{ax}}(\mathcal{M}_{-k})\}^2, \quad (5)$$

where I_k is the k th fold and $\mathcal{M}_{-k}$ is fit without that fold. In the code, $K = \min(10, n)$ and the folds are shuffled with random seed 42. Cross-validation is used here as an estimate of predictive risk; Arlot and Celisse review this risk-estimation and model-selection interpretation in Sections 1, 4, and 6 of their survey [1].

4 The Parametric Spiral Models

The parametric family used in the current `src/cleaned_linefield_comparison.py` experiment is a one-angle vector-field basis. After centering the cleaned coordinates, write

$$r_i = \sqrt{x_i^2 + y_i^2}, \quad \theta_i = \text{atan2}(y_i, x_i).$$

For exponent p and pitch parameter γ , the implemented vector has radial component $\cos \gamma$ and tangential component $-\sin \gamma r_i^p$. Therefore the predicted line angle is

$$\hat{\alpha}_{p,\gamma}(z_i) = \theta_i + \text{atan2}(-\sin \gamma r_i^p, \cos \gamma) \mod \pi. \quad (6)$$

This formula is the mathematical model being tested. The fixed families are $p = 0$ (logarithmic label), $p = 1$ (Archimedean label), and $p = 2$ (Fermat label). The continuous parametric family fits $p \in (-1, 3)$. For fixed p , the code searches over $\gamma \in [-\pi, \pi]$ and then refines the best values with L-BFGS-B. For the continuous family, the code searches a grid over p and γ and then optimizes both parameters.

Equation (6) is deliberately much smaller than a kernel model. It asks whether one global spiral exponent and one pitch angle can describe the whole cleaned field. This is an interpretable hypothesis, but it is also restrictive: if the data contain local bends, local noise, or spatially varying pitch, a single pair (p, γ) cannot adapt to them.

5 The Kernel Models

The kernel models first convert each observed axial angle to the doubled-angle vector

$$u_i = (\cos 2\hat{\alpha}_i, \sin 2\hat{\alpha}_i) \in \mathbb{R}^2. \quad (7)$$

This maps each line to a point on the unit circle without distinguishing $\hat{\alpha}_i$ from $\hat{\alpha}_i + \pi$. A Gaussian process is fit independently to the two coordinates of u_i using the same covariance matrix. If the predicted vector at z is $\hat{u}(z) = (\hat{u}_1(z), \hat{u}_2(z))$, the predicted line angle is

$$\hat{\alpha}(z) = \frac{1}{2} \text{atan2}\{\hat{u}_2(z), \hat{u}_1(z)\}. \quad (8)$$

The three exact kernels in the experiment are

$$K_{\text{RBF}}(z, z'; \ell) = \exp\left[-\frac{\|z - z'\|_2^2}{2\ell^2}\right], \quad (9)$$

$$K_{\text{unif}}(z, z'; \ell) = \mathbf{1}\{\|z - z'\|_2 \leq \ell\}, \quad (10)$$

$$K_{\text{mult}}(z, z'; \ell, \kappa) = \exp\left[-\frac{\|z - z'\|_2^2}{2\ell^2}\right] \exp[\kappa \cos\{\vartheta(z) - \vartheta(z')\}], \quad (11)$$

where $\vartheta(z) = \text{atan2}(y, x)$. The Gaussian RBF kernel says that nearby points should have similar line directions. The uniform kernel is a local-neighbourhood baseline. The multiplicative kernel says that two points are especially related when they are both spatially close and have similar polar angle around the centre.

The length-scale grid is not the old absolute grid `geomspace(0.2, 220, 14)`. It is now dataset-adaptive. Let

$$s_X = \max\{\text{range}(x), \text{range}(y), 1\}, \quad d_{\text{nn}} = \text{median}_i \min_{j \neq i} \|z_i - z_j\|_2.$$

The code sets

$$\ell_{\min} = \max\left\{\frac{d_{\text{nn}}}{2}, \frac{s_X}{80}, 10^{-3}\right\}, \quad \ell_{\max} = \max\{1.5s_X, 4\ell_{\min}\}, \quad (12)$$

and then uses 14 geometrically spaced values

$$\ell_j = \ell_{\min} \left(\frac{\ell_{\max}}{\ell_{\min}}\right)^{j/13}, \quad j = 0, \dots, 13. \quad (13)$$

This is what **geomspace** means: adjacent candidate values have the same multiplicative ratio, so the sweep covers small, medium, and large spatial scales evenly on a logarithmic scale. The multiplicative kernel additionally sweeps

$$\kappa \in \{0, 0.5, 1, 2, 4, 8, 16\}.$$

For every kernel candidate, the observation-noise parameter σ_n is also swept over eight geometrically spaced values,

$$\sigma_n \in \text{geomspace}(0.05b, 0.9b, 8), \quad b = \max\left\{\frac{\text{sd}(u_1) + \text{sd}(u_2)}{2}, 0.05\right\}. \quad (14)$$

Thus the kernel comparison is not using one fixed noise value across all datasets.

6 Universality and What It Means Here

Universality is an approximation property, not a score. A kernel is universal on a compact domain X when its RKHS is dense in $C(X)$, the continuous real-valued functions on X under the uniform norm. In plain language: with enough data and the right regularization, functions in the RKHS can approximate any continuous target pattern on that compact domain.

The Gaussian RBF kernel is universal on compact subsets of Euclidean space. The precise reference is Steinwart [6]: Definition 4 introduces universal kernels, Corollaries 10 and 11 give universality conditions using Taylor or Fourier expansions, and Example 1 states the Gaussian RBF case. This theorem supports the use of the RBF model as a flexible non-parametric family. It does not say that the RBF will automatically win finite-sample prediction, and it does not remove the need to tune ℓ and σ_n .

The multiplicative kernel in (11) inherits the RBF's spatial flexibility and multiplies it by a von-Mises angular similarity. The angular factor has the Fourier expansion

$$\exp\{\kappa \cos(\vartheta - \vartheta')\} = I_0(\kappa) + 2 \sum_{m=1}^{\infty} I_m(\kappa) \cos\{m(\vartheta - \vartheta')\},$$

where $I_m(\kappa) > 0$ for $\kappa > 0$. The positive Fourier coefficients are the mathematical reason this factor can represent smooth angular variation around the centre. The uniform kernel is different: it is included as a local averaging baseline, not because we rely on a universality theorem for it.

7 Evidence, Bayes Factors, and BIC

RSS asks, ‘‘How well did the fitted model predict held-out angles?’’ Evidence asks a different question: ‘‘How probable are the observed responses after integrating over the latent function values allowed by the model?’’ For a kernel with covariance matrix

$$C_\theta = K_\theta(X, X) + \sigma_n^2 I_n, \quad (15)$$

and doubled-angle response matrix $U \in \mathbb{R}^{n \times 2}$, the implemented Gaussian-process log marginal likelihood is

$$\log p(U \mid X, \theta) = -\frac{1}{2} \text{tr}(U^\top C_\theta^{-1} U) - \log |C_\theta| - n \log(2\pi). \quad (16)$$

The first term rewards fit. The log-determinant term penalizes models that spread probability mass too diffusely over many possible response patterns. This is the Gaussian-process Occam factor. Rasmussen and Williams derive the one-output version in *Gaussian Processes for Machine Learning*, Chapter 2, Equation (2.30), and Algorithm 2.1 gives the Cholesky implementation used by standard GP code [4]. Equation (16) is the two-output independent-coordinate extension of that formula.

For the finite-dimensional parametric models, the experiment uses the BIC approximation

$$\text{BIC}(\mathcal{M}) = k_{\mathcal{M}} \log n - 2 \log L(\hat{\theta}_{\mathcal{M}}), \quad \log p(y \mid \mathcal{M}) \approx -\frac{1}{2} \text{BIC}(\mathcal{M}). \quad (17)$$

Here $k_{\mathcal{M}} = 1$ for fixed p because only γ is fit, and $k_{\mathcal{M}} = 2$ for continuous p because both p and γ are fit. Schwarz introduced this large-sample criterion in *The Annals of Statistics*, 6(2):461–464 [5]. Kass and Raftery, Section 3 and their opening summary, explain Bayes factors and note that BIC gives a rough approximation to log Bayes factors [2].

Across datasets, the total evidence for model m is

$$E_m^{\text{tot}} = \sum_{j=1}^{39} E_{mj}. \quad (18)$$

The Bayes factor comparing model m to the best model $m_{\star}$ is

$$\text{BF}_{m,m_{\star}} = \exp\{E_m^{\text{tot}} - E_{m_{\star}}^{\text{tot}}\}, \quad \log_{10} \text{BF}_{m,m_{\star}} = \frac{E_m^{\text{tot}} - E_{m_{\star}}^{\text{tot}}}{\log 10}. \quad (19)$$

This is why the Bayes factor is related to the residual score but is not the same number. Residuals enter the likelihood term, but the evidence also contains covariance geometry, noise scale, and the log-determinant complexity penalty.

8 Results on the Cleaned Datasets

Table 1 reports the held-out predictive experiment. The multiplicative RBF–von-Mises RKHS has the smallest mean test RSS, the smallest median test RSS, and the smallest mean absolute axial error. It also wins the dataset-by-dataset RSS comparison on 32 of 39 datasets; the uniform kernel wins 5, and the Gaussian RBF wins 2.

Model	Mean test RSS	Median test RSS	Mean MAE
Multiplicative RBF–VM RKHS	0.1079	0.1060	11.50°
Gaussian RBF RKHS	0.1291	0.1160	12.54°
Uniform RKHS	0.1302	0.1176	12.81°
Parametric $p = 1$ (Archimedean)	0.6783	0.6742	39.88°
Parametric $p = 2$ (Fermat)	0.6828	0.6670	40.11°
Parametric $p = 0$ (Logarithmic)	0.6889	0.6875	40.14°
Parametric $p \in (-1, 3)$	0.6953	0.6789	40.33°

Table 1: Held-out axial prediction on 39 cleaned datasets. Lower RSS and lower MAE are better.

Table 2 reports the evidence-style comparison. The same multiplicative kernel is the aggregate winner. The ordering is not identical to RSS because evidence is not a held-out loss: it rewards fit, penalizes unnecessary flexibility through the log determinant, and depends on the optimized noise scale. In dataset-by-dataset evidence winners, the multiplicative kernel wins 17 of 39 datasets, parametric $p = 0$ wins 12, parametric $p = 2$ wins 5, parametric $p = 1$ wins 4, and the Gaussian RBF wins 1. Aggregating evidence over all datasets still favours the multiplicative kernel by a large margin.

Model	Mean evidence	Total evidence	$\log_{10} \text{BF}$
Multiplicative RBF–VM RKHS	−137.867	−5376.807	0.000
Parametric $p = 1$ (Archimedean)	−148.523	−5792.401	−180.490
Parametric $p = 2$ (Fermat)	−148.781	−5802.467	−184.862
Parametric $p = 0$ (Logarithmic)	−148.951	−5809.108	−187.746
Parametric $p \in (-1, 3)$	−149.981	−5849.265	−205.186
Gaussian RBF RKHS	−152.740	−5956.851	−251.910
Uniform RKHS	−261.668	−10205.062	−2096.884

Table 2: Evidence-style comparison on the same 39 cleaned datasets. Kernel rows use optimized GP marginal likelihood for the doubled-angle embedding; parametric rows use $-\text{BIC}/2$ from axial residuals. Bayes factors are relative to the multiplicative RBF–VM RKHS model.

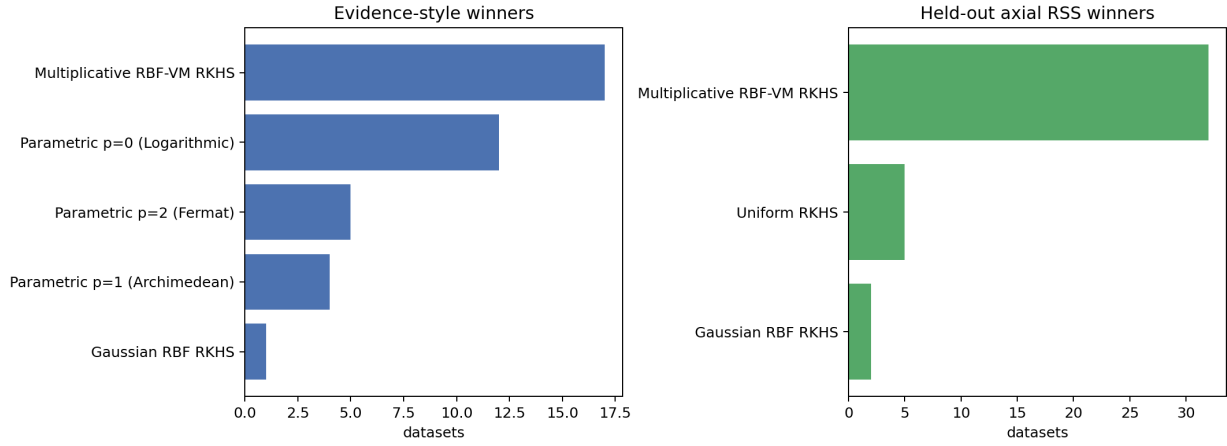


Figure 3: Dataset-by-dataset winners for the evidence-style score and held-out axial RSS. This plot explains why the aggregate evidence and RSS tables should not be read as identical experiments: RSS chooses the model with lowest held-out prediction error, while evidence also includes a covariance-volume penalty.

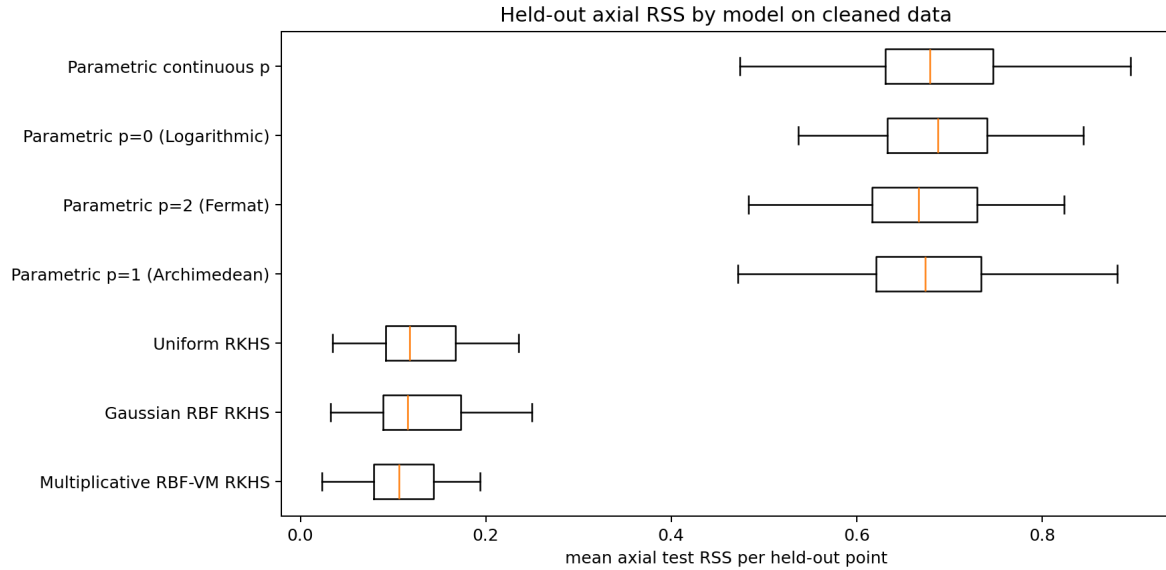


Figure 4: Distribution of held-out axial RSS across the 39 cleaned datasets. The vertical ordering matches Table 1: the multiplicative RBF-VM kernel has the lowest typical predictive error, followed by the Gaussian and uniform kernels, while the current parametric spiral basis has much larger held-out axial error.

9 Streamline Diagnostics

The numerical scores are the primary evidence, but streamline plots help explain what the fitted line fields look like. A streamline is an integral curve of the fitted orientation field. If $\hat{\alpha}(z)$ is the fitted line angle, the displayed curves follow the local direction

$$\frac{dz(t)}{dt} = \{\cos \hat{\alpha}(z(t)), \sin \hat{\alpha}(z(t))\},$$

with the understanding that the opposite sign gives the same line field. These plots are diagnostic rather than part of the scoring function: the scores come from Equations (5) and (16), while the streamlines translate the fitted angle functions into geometry.

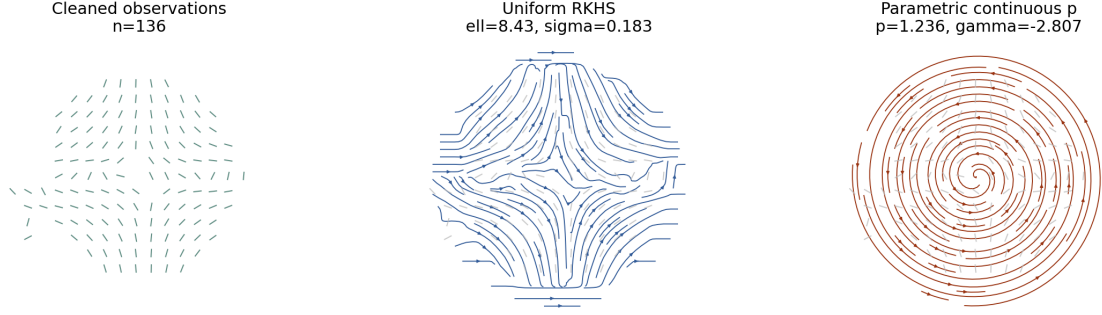


Figure 5: Fitted streamlines for one cleaned dataset. The diagnostic question is whether the fitted field follows the same local circular flow as the observed line segments. Kernel smoothers can bend locally because $\hat{\alpha}(z)$ is learned through $K(z, z')$, whereas the parametric family must express the field through the single global form in Equation (6).

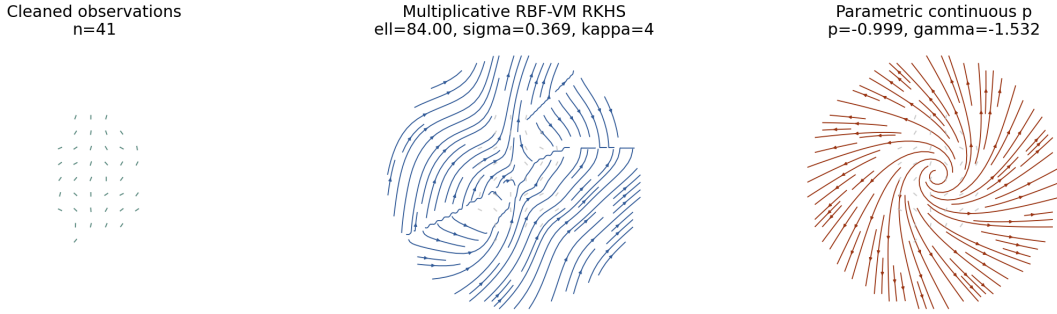


Figure 6: A second fitted-streamline example from a dataset with heavier cleaning. The visual comparison should be read together with the raw-clean study in Figure 1: when peripheral segments are removed, the fitted field is evaluated on the central region where the circular line pattern is the object of study.

The consulting interpretation is therefore simple but important. The cleaned central field is not best described by the current one-exponent parametric spiral family. The parametric models are interpretable and sometimes score well under evidence because they are low-dimensional, but their held-out errors are much larger. The multiplicative kernel performs best because it has the right kind of flexibility: it is smooth in Euclidean location and also respects polar position around the centre.

10 What Is Certain, and What Is Not

The exact data cleaning rule, axial residual, parametric formula, kernels, hyperparameter grids, RSS table, evidence table, and Bayes-factor table are backed by the cleaned-experiment code and its generated CSV files. The relevant local paths are `src/cleaned_linefield_comparison.py` and `data/cleaned_linefield_comparison/`. The citations above back the statistical tools: axial doubling, cross-validation, GP marginal likelihood, BIC, Bayes factors, and RBF universality.

Two points are deliberately stated as limitations rather than conclusions. First, the ROI cleaning rule is an experimental design choice, not a statistical theorem. Second, the evidence table mixes exact GP marginal likelihoods for the kernel models with BIC approximations for the parametric models. That is a useful evidence-style comparison, but it is not the same as a fully Bayesian analysis with matched priors over every kernel and parametric hyperparameter.

11 Citation Map

Concept used here	Where to find it in the literature
Axial angles and doubled angles	Mardia and Jupp, <i>Directional Statistics</i> , Chapter 1, Section 1.1, where axial data are converted to circular data by doubling angles [3].
Delta-method intuition for transformed estimators	van der Vaart, <i>Asymptotic Statistics</i> , Chapter 3, especially Theorem 3.1 on the delta method [7].
Cross-validation as risk and model-selection tool	Arlot and Celisse, <i>Statistics Surveys</i> 4:40–79, Sections 1, 4, and 6 [1].
GP marginal likelihood	Rasmussen and Williams, <i>Gaussian Processes for Machine Learning</i> , Chapter 2, Equation (2.30), and Algorithm 2.1 [4].
Universality of Gaussian RBF kernels	Steinwart, <i>JMLR</i> 2:67–93, Definition 4, Corollaries 10–11, and Example 1 [6].
BIC/Schwarz criterion	Schwarz, <i>The Annals of Statistics</i> 6(2):461–464 [5].
Bayes factors and BIC approximation	Kass and Raftery, <i>JASA</i> 90(430):773–795, especially Section 3 and the opening summary comments on BIC [2].

References

- [1] Sylvain Arlot and Alain Celisse. A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4:40–79, 2010.
- [2] Robert E. Kass and Adrian E. Raftery. Bayes factors. *Journal of the American Statistical Association*, 90(430):773–795, 1995.
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- [5] Gideon Schwarz. Estimating the dimension of a model. *The Annals of Statistics*, 6(2):461–464, 1978.
- [6] Ingo Steinwart. On the influence of the kernel on the consistency of support vector machines. *Journal of Machine Learning Research*, 2:67–93, 2001.
- [7] Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, 2000.